

1. Administrative & Study Identification

Full Study Title: An open-label, multi-center, phase 2 basket study to assess efficacy, safety and pharmacokinetics of iptacopan (LNP023) in participants with autoimmune benign hematological disorders **Short Title/Acronym:** LNP023 Basket Study in Autoimmune Benign Hematological Disorders **Protocol Number:** CLNP023L12201 **NCT Number:** NCT05086744 **Posting ID:** 655081352049538703 **Sponsor/Company Name:** Novartis Pharmaceuticals / Novartis Pharma AG **Investigational Product:** Iptacopan hydrochloride (LNP023) **Phase of Development:** Phase II **Trial Status:** Terminated (Study was terminated early) **Medical Monitor:** Novartis Clinical Operations Lead / Nicola Cooper (Lead Contact)

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To assess the ability of iptacopan to induce a clinically meaningful response, defined as an increase in platelet count for primary Immune Thrombocytopenia (ITP) or an increase in hemoglobin levels for primary Cold Agglutinin Disease (CAD). **Secondary Objectives:** To evaluate time to first response, duration of response, magnitude of response, relevant disease biomarker levels, pharmacokinetics (PK), safety, and tolerability (including FACIT-Fatigue).

2.2 Background & Rationale Immune thrombocytopenia (ITP) and cold agglutinin disease (CAD) are autoimmune hematologic disorders causing thrombocytopenia and hemolytic anemia, respectively. ITP is defined clinically as thrombocytopenia that results from immune destruction of platelets. While there are existing therapies, many patients remain refractory or experience relapse, and there is a notable lack of effective oral treatments for CAD. Iptacopan hydrochloride is a first-in-class, oral, selective inhibitor of factor B (a key component of the complement alternative pathway). This basket study addresses the unmet medical need by evaluating the efficacy and safety of iptacopan monotherapy in adult patients with primary ITP or primary CAD who have failed at least one prior unique therapy.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional (Basket study)
Control Type: Single-arm (within each indication cohort)
Blinding: Open-label **Randomization:** Non-randomized

3.2 Planned Enrollment & Duration Targeted Enrollment: 19 patients total (9 ITP, 10 CAD) **Number of Sites:** 8 centers across 6 countries (Germany, Italy, South Korea, Spain, United Kingdom, and USA) **Estimated Study Start:** 21-Dec-2021 **Study Status Description:** Study was terminated early.

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria Criteria Type: PHASE_REQUIREMENT (Trial must be in PHASE2) 1. Adult participants weighing at least 35 kg. 2. **Cohort 1 (ITP):** Diagnosis of persistent or chronic primary ITP (diagnosed ≥ 3 months prior) with sustained thrombocytopenia (platelets $< 30 \times 10^9/L$) after receiving ≥ 1 unique prior ITP-directed therapy. 3. **Cohort 2 (CAD):** Diagnosis of primary CAD with laboratory evidence of ongoing hemolysis and sustained anemia (hemoglobin < 10 g/dL) after receiving ≥ 1 unique prior CAD-directed therapy.

4.2 Exclusion Criteria 1. Known or suspected hereditary or acquired complement deficiency, or history of primary/secondary immunodeficiency (including positive HIV, chronic Hepatitis B or C). 2. History of recurrent invasive infections caused by encapsulated organisms (Neisseria meningitidis, Streptococcus pneumoniae, Haemophilus influenzae). 3. Secondary ITP or secondary cold agglutinin syndrome. No CAD-directed background therapy is permitted.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Iptacopan hydrochloride 200 mg twice daily (b.i.d.). **Route of Administration:** Oral. **Duration of Treatment:** A 12-week core treatment period (Part A). Responders have the option to enter an extended treatment period for up to 24 months (Part B), making the maximum duration up to 31 months.

5.2 Concomitant & Prohibited Therapy Permitted: For ITP, either a thrombopoietin receptor agonist or low-dose corticosteroid is permitted only if kept at a stable dosage for at least 4 weeks prior to the first iptacopan dose. **Prohibited:** Concurrent CAD-directed background therapy. Use of other investigational drugs within 5 half-lives or 30 days of enrollment.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Up to Day 0	Informed Consent, baseline hematology, complement activation stratification (sC5b-9 level), mandated vaccinations for encapsulated organisms.
Baseline	Day 0	First dose of iptacopan hydrochloride 200 mg

b.i.d. | | **Part A (Core)** | Weeks 1 to 12 | Efficacy assessment (Platelet/Hb tracking), safety labs, PK monitoring. | | **Washout / Follow-Up** | ≤ 4 Weeks (Post-Part A) | Washout for responders before extension, or safety follow-up for non-responders. | | **Part B (Extension)** | Up to 24 Months | Extended safety and durability of response evaluation for eligible responders. |

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Clinically meaningful response during the 12-week treatment period (Part A). * *For ITP:* A platelet count of $\geq 50 \times 10^9/L$ sustained for at least two consecutive weeks without the use of rescue therapy. * *For CAD:* A hemoglobin (Hb) level increase of ≥ 1.5 g/dL above baseline sustained for at least two consecutive weeks without the use of rescue therapy. **Measurement Method:** Central/Local Laboratory Analysis.

7.2 Secondary & Exploratory Endpoints * Time to first response (Time to first platelet count $\geq 50 \times 10^9/L$ for ITP; Time to first Hb level ≥ 1.5 g/dL above baseline for CAD). * Duration of response (calculated cumulatively if there are non-continuous periods of response, tracking the duration of time platelet counts remain $\geq 50 \times 10^9/L$ without rescue therapy). * Magnitude of response and changes in relevant disease biomarker levels. * Patient-reported outcomes including the FACIT-Fatigue scale.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard continuous collection throughout the trial with a particular focus on infection risks due to alternative complement pathway inhibition. Prophylactic vaccination was strictly required. **Serious Adverse Events (SAEs):** Standard 24-hour reporting protocols for severe or life-threatening occurrences. **Data Monitoring Committee (DMC):** Internal Novartis clinical trial safety oversight.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Efficacy and safety analyses conducted on all enrolled patients who received at least one dose of the study drug. **Sample Size Justification:** As a Phase 2, open-label, non-confirmatory basket study, the sample size was selected to evaluate preliminary efficacy and safety signals within distinct rare disease cohorts rather than strictly powered hypothesis testing. **Handling of Missing Data:** Duration of response was explicitly designed to be analyzed cumulatively for responders, factoring in potential non-continuous response windows without directly imputing missing gaps as failures.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study was conducted in accordance with Good Clinical Practice (GCP) guidelines and the Declaration of Helsinki. **Monitoring Plan:** Overseen via the Novartis Quality Management System (QMS), encompassing all clinical quality assurance and quality control activities. **Audit Trail:** eCRF locking, strict protocol amendment controls, and anonymized patient-level data sharing via the ClinicalStudyDataRequest portal.

11. Study Identifiers & Governance

Primary ID: CLNP023L12201 **Registry Number:** NCT05086744 **Posting ID:** 655081352049538703 **Sponsor/Lead Organization:** Novartis **Study Title:** Basket Study to Assess Efficacy, Safety and PK of Iptacopan (LNP023) in Autoimmune Benign Hematological Disorders **Official Regulatory Title:** An open-label, multi-center, phase 2 basket study to assess efficacy, safety and pharmacokinetics of iptacopan (LNP023) in participants with autoimmune benign hematological disorders

12. Clinical Framework (Autoimmune Hematology Specific)

Disease Areas: Immune Thrombocytopenia (ITP), Cold Agglutinin Disease (CAD) **Condition Name:** Immune Thrombocytopenia (ITP) **Preferred UMLS Name:** Autoimmune Hematologic Disease **Preferred Name:** Thrombocytopenia due to platelet alloimmunization **Semantic Type:** Disease or Syndrome **MeSH Code:** D016833 **SNOMED ID:** 2897005 **Diagnosis Threshold:** Platelets $\leq 30 \times 10^9/L$ (ITP); Hemoglobin ≤ 10 g/dL (CAD) **Medical Methodology:** Selective Factor B (Alternative Complement Pathway) Inhibition **Optimization Target:** Sustained Platelet count $\geq 50 \times 10^9/L$ or Hemoglobin increase ≥ 1.5 g/dL

13. Study Procedures & Methodology

Management Pathway: Targeted oral complement inhibition (iptacopan hydrochloride monotherapy) **Education Protocol (HCP):** Mandated vaccination guidelines against encapsulated bacteria (N. meningitidis, S. pneumoniae, H. influenzae) **Education Protocol (Patient):** Infection risk awareness and continuous monitoring adherence **Quality Audit Mechanism:** Novartis global Quality Management System (QMS) standard auditing

14. Observation & Follow-Up Schedule

Total Duration: Up to 31 Months (for responders proceeding to Part B) **Onsite Visit Cadence:** Initial 12-week core phase, followed by 4-week washout, progressing into a 24-month extension phase **Remote Monitoring Frequency:** Routine safety and hematology laboratory assessments as per study protocol **Checklist Review Interval:** 12-week core milestone for primary endpoint achievement

15. Sign-off & Approval

Role	Name	Signature	Date
Principal Investigator	Nicola Cooper	_____	03-Apr-2026
Clinical Operations Lead	Novartis Representative	_____	03-Apr-2026
Lead Statistician	Novartis Biostatistics	_____	03-Apr-2026